AUTONOMOUS AI SWARM | HOSTILE DATA RESILIENT | DEFENSE & CRISIS COORDINATION

RESCUESWARM AI

"When everything goes wrong, the swarm reorganizes."

GITHUB REPOSITORY
subwaycookiecrunch/RescueSwarm-AI

TEAM & ROLES
Daksh (Backend) • Raj (AI & Swarm) • Somesh (Domain & Pitch)

STATUS
Full Stack Live (FastAPI + React + WebSockets)

⚡ 1. The Real-World Crisis Problem

Why Normal Emergency Software Fails During Large Catastrophes

During disasters (landslides, flash floods, quakes, urban structural collapses), crisis centers aren't bottlenecked by lack of data. They are overwhelmed by a **hostile information environment**:

- **Incomplete & Contradictory:** One caller reports 5 casualties; panic tweets claim 500 dead.
- **Duplicated & Stale:** 20 calls describe the same building collapse from 25 minutes ago.
- **Manipulated / Fake Rumors:** Social media bots inject dramatic fake distress alerts that drain scarce units.
- **Rapid Terrain Shifting:** Key bridges and roads collapse mid-transit, invalidating human dispatcher plans.

🏛️ 2. The Solution: RescueSwarm Multi-Agent Swarm

RescueSwarm converts chaotic, hostile crisis streams into synchronized, ground-truth physical emergency actions via an autonomous 5-agent coordination architecture:

SWARM AGENT	CORE FUNCTION	UNDERLYING INTELLIGENCE ENGINE
1. Verification Agent Noise Defense	Filters adversarial rumors, deduplicates repeated citizen calls, correlates multi-source evidence.	Spatial clustering (<800m), temporal windowing (<15m), and Bayesian source weighting (Sensors: 0.95, Responders: 0.98, Social: 0.25).
2. Triage Agent Severity Matrix	Assesses verified casualties, structural hazards, and determines crisis response priorities.	Dynamic priority queue classifying scenes into `CRITICAL — MULTIPLE CASUALTIES`, `HIGH`, `MEDIUM`, or `LOW`.
3. Resource Agent Fleet Dispatch	Assigns nearest qualified emergency vehicles (ALS Ambulances, USAR Rescue Teams, Fire Tenders).	Real-time nearest-neighbor fleet allocation matrix matching vehicle medical capability to scene severity.
4. Routing Agent Dynamic Graph	Calculates optimal navigation across road graph. Reacts to landslides by computing detours.	NetworkX Dijkstra/A* graph algorithm with dynamic edge invalidation and real-time rerouting around hazards.
5. Recovery Agent Self-Healing	Heartbeat watchdog. Detects node deaths, transfers roles, and syncs state to standby nodes.	400ms heartbeat polling. Detects silence >1.4s, executes automated state failover with zero packet loss.

3. The Two Core Differentiators

★ Differentiator 1: Self-Healing AI Swarm

Normal multi-agent architectures are brittle: if the Routing Agent process crashes, all navigation halts.

```
Routing Agent Crashes 💥
  ↓ Heartbeat silence > 1.4s
Recovery Watchdog Detects Failure
  ↓ Responsibilities redistributed
Backup Standby Node Promoted
  ↓ Active routes & graph synchronized
Swarm Restored (Health: 100%) ✓
```

★ Differentiator 2: Hostile Data Immunity

When 47 chaotic reports hit the system, RescueSwarm cross-validates them against physical IoT sensors and responder ground units.

```
Injected: "100 TRAPPED AT LOCATION X"
Source: Anonymous Telegram post
Sensor Corroboration: 0
Responder Radio: 0
Calculated Confidence: 23%
STATUS: QUARANTINED (No units wasted)
```

4. The Finale Pitch Demonstration ("Try to Break It")

Rather than a passive presentation, the team challenges the judges: *"Instead of showing you RescueSwarm, we're going to try to break it."*

`00:00`	**Baseline Normal City:** 0 Incidents, 5/5 Agents Active, Cluster Health: 100%. Road network clear.
`00:10`	**Disaster Wave Injected:** 47 chaotic incoming reports hit the stream. Verification Agent clusters them into 4 real incidents with 96% confidence.
`00:20`	**Triage & Dispatch:** Mall Road classified as `CRITICAL — MULTIPLE CASUALTIES`. Resource Agent assigns `Ambulance-03`; Routing Agent plots route through Aleo Bridge (Road B).
`00:30`	**Chaos #1 — BLOCK ROAD B:** Operator triggers road collapse. Active route invalidated immediately; swarm calculates dynamic detour around river.
`00:40`	**Chaos #2 — HOSTILE FAKE NEWS:** Operator injects *"100 PEOPLE TRAPPED AT LOCATION X"*. Verification Agent calculates 23% confidence and quarantines the claim. Zero units wasted.
`00:50`	**Chaos #3 — KILL ROUTING AGENT:** Hard process termination. Heartbeat ceases. Recovery Watchdog detects failure within 1.4s, transfers authority to Backup Node, and restores health to 100%.

5. What This Project Is NOT

Strict Architectural Principles:

- ❌ Not a disaster prediction tool
- ❌ Not a chatbot pretending to be agents
- ❌ Not a static non-interactive map
- ❌ Not generic emergency reporting
- ✅ Real event-driven asynchronous pub/sub
- ✅ Deterministic NetworkX pathfinding
- ✅ Bayesian sensor correlation matrix
- ✅ Live WebSocket command center dashboard